

CBSE Class 11 & 12 Computer Science (083)

Unit III: Standard Python Libraries & Mathematical Modules

Authoritative Syllabus & Laboratory Practical Guidelines

1. Overview & Import Syntax

Python provides the built-in "math" module that conforms to the C standard for mathematical functions. Unlike basic arithmetic operators (+, -, *, /), advanced scientific and algebraic calculations require importing the module: "import math" or "from math import ceil, floor, sqrt, pi". All return values from math functions are floating-point numbers unless specified otherwise.

2. Mathematical Constants

- `math.pi`: Mathematical constant π (ratio of circumference to diameter)
- `math.e`: Euler's number $e = 2.718281828459045$ (base of natural logarithms)
- `math.tau`: Circle constant $2\pi = 6.283185307179586$
- `math.inf`: Positive floating-point infinity (equivalent to `float("inf")`)
- `math.nan`: Floating-point "Not a Number" (NaN) value for undefined operations

3. Number-Theoretic & Representation Functions

- `math.ceil(x)`: Returns the smallest integer greater than or equal to x (e.g. `math.ceil(4.2)` -> 5).
- `math.floor(x)`: Returns the largest integer less than or equal to x (e.g. `math.floor(4.9)` -> 4).
- `math.trunc(x)`: Returns the truncated integer value of x , removing all fractional digits towards zero.
- `math.fabs(x)`: Returns the absolute value of x as a float (difference from `abs()` which preserves int type).
- `math.factorial(n)`: Returns $n!$ for non-negative integers (raises `ValueError` for negative numbers or floats).
- `math.gcd(*integers)`: Returns the greatest common divisor of specified integer arguments.
- `math.lcm(*integers)`: Returns the least common multiple of specified integer arguments.

4. Power, Root & Logarithmic Operations

- `math.sqrt(x)`: Returns the square root of x . Raises `ValueError` if $x < 0$.
- `math.isqrt(n)`: Returns the exact integer square root of a non-negative integer n .
- `math.pow(x, y)`: Returns x raised to power y as a float (differs from `x ** y` which preserves integers).
- `math.exp(x)`: Returns e raised to the power x (e^x).
- `math.log(x, [base])`: Returns natural logarithm $\ln(x)$ if base omitted, or log to the specified base.
- `math.log10(x)`: Returns the base-10 logarithm of x .

5. Trigonometry & Angular Conversions

- Angles in all math trigonometric functions (`sin`, `cos`, `tan`) MUST be expressed in RADIANS, not degrees.
- `math.radians(deg)`: Converts angle degrees to radians (e.g. `math.radians(180)` -> 3.141592653589793).
- `math.degrees(rad)`: Converts angle radians to degrees (e.g. `math.degrees(math.pi)` -> 180.0).
- `math.hypot(*coordinates)`: Returns the Euclidean norm / hypotenuse $\sqrt{x^2 + y^2}$ from the origin.

6. Mandatory CBSE Laboratory Practical Exercises

Exercise 1 (Trigonometric Height Calculation): Write a Python program to compute the height of a tower given the distance from the observer and the angle of elevation in degrees using `math.tan()` and `math.radians()`.

Exercise 2 (Euclidean Hypotenuse & Coordinates): Calculate the straight-line distance between two 2D points (x_1, y_1) and (x_2, y_2) using `math.hypot(x2 - x1, y2 - y1)`.

Exercise 3 (Mathematical Assertion & Tracing): Understand the difference between `math.floor(-3.2) = -4` vs `math.trunc(-3.2) = -3`.